# CSE 330-Machine Learning Project Proposal

## A Comparative Study of Classical and Quantum Machine Learning Approaches on Clinical Datasets

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# 1

# Introduction

# Introduction

Our team, **Human Machines**, seeks to leverage state-of-the-art quantum computing technology by integrating its advantageous properties into classical ML-DL frameworks, with the objective of enhancing computational speed, reducing costs, and improving accuracy.

The primary goal of this term project is to establish comprehensive **benchmarks** comparing classical models, existing quantum models, and our proposed novel quantum architectures.This research will address existing gaps in the literature where certain model architectures have not been explored within quantum computing frameworks.

# Introduction

Our ultimate objective is to develop this research for **publication** in a peer-reviewed venue, contributing to both quantum computing and healthcare informatics communities.

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# 2

# Motivation

# Motivation

- Quantum computing is a rapidly growing field that has received a significant amount of support in the past decade in industry and academia[1].

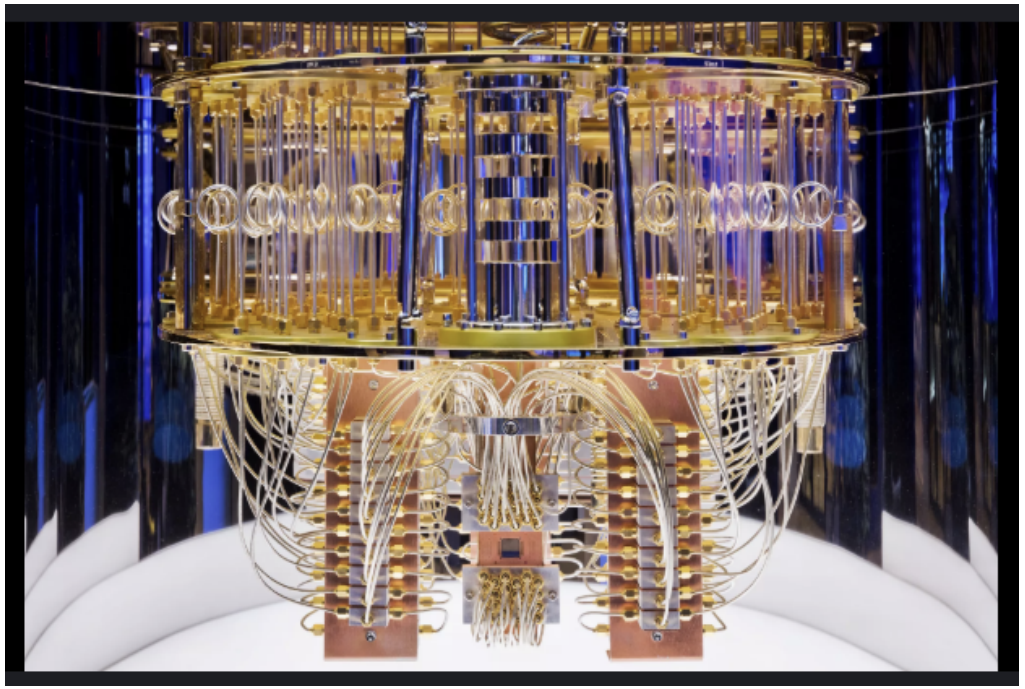


**Figure 1:** Noisy Intermediate Scale Quantum Computer deployed by IBM

# Motivation

- Several physical quantum computers are now freely available to use through cloud services, with some implementations supporting upwards of hundreds of qubits. These advances mark the beginning of the Noisy Intermediate-Scale Quantum (NISQ) era of quantum computing, paving the way for hybrid quantum-classical systems[1].
- Quantum machine learning represents a highly promising realm in contemporary physics and computer science research, with far-reaching implications spanning quantum chemistry, artificial intelligence, and even high-energy physics. Nevertheless, it remains in its nascent stages of development[2].

# Motivation

- Fuelled by increasing computer power and algorithmic advances, machine learning techniques have become powerful tools for finding patterns in data. Since quantum systems produce counter-intuitive patterns believed not to be efficiently produced by classical systems, it is reasonable to postulate that quantum computers may outperform classical computers on machine learning tasks[3].
- For example, Classical AI methods (ML, DL) face limitations in managing high-dimensional vector spaces and modeling complex gene relationships, including challenges with hyperparameter tuning, computational costs, and processing power requirements[4].

# Motivation

- Quantum machine learning models can naturally capture complex, high-order relationships among many features by encoding data into high-dimensional quantum states. Using quantum feature maps, entanglement, and interference, these models can represent intricate feature dependencies that are difficult for classical models to learn efficiently, especially in high-dimensional datasets. [5]

- Considering the potential of quantum machine learning, after conducting a literature review, we found it interesting to benchmark clinical datasets (primarily tabular) using classical models, **existing quantum models**, and **our proposed quantum models** that have not been trained with quantum approaches before. This aims to experiment with quantum advantages, fill the literature gap, and show future directions.

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# Problem Definition

# Problem Definition

**Primary Research Questions:**

1. Given clinical datasets for classification tasks, which modeling approach demonstrates superior performance: classical machine learning models, existing quantum machine learning models[2][1], or our proposed custom-modified quantum architectures using NISQ[2] devices?
2. Is there empirical evidence for NISQ devices to outperform classical methods in the absence of FTASQ (Fault-Tolerant Application-level Quantum) devices?
3. Do NISQ devices demonstrate the feasibility and potential to significantly outperform classical methods when scaled to FTASQ devices?

# Problem Definition

**Performance Evaluation Criteria:**

- Classification accuracy and precision
- Training time and computational efficiency
- Model generalizability across datasets
- Scalability with respect to data dimensions
- Resource requirements (qubits, classical compute)
- Quantum advantage indicators for future FTASQ scaling

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# Existing Works

# Existing Works

1. **Quantum-enhanced classification of brain tumors using DNA microarray gene expression profile [4]**

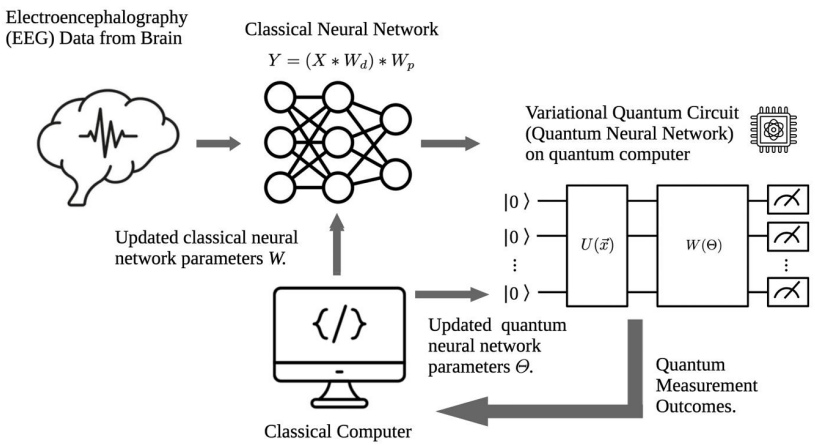
TABLE I: Accuracy scores of the proposed Deep VQC, and various ML algorithms reported in the CuMiDa database

| Classifier | Accuracy Score |
|---|---|
| Deep VQC (Ours) | 0.85 |
| Decision Tree (DT) | 0.85 |
| Hoeffding Tree (HC) | 0.38 |
| Naive Bayes (NB) | 0.85 |
| Random Forest (RF) | 0.91 |
| K-Nearest Neighbors (KNN) | 0.87 |
| Multi-Layer Perceptron (MLP) | 0.82 |
| Support Vector Machine (SVM) | 0.95 |
| ZeroR | 0.35 |
| k-Means | 0.46 |

**Figure 2:** Deep VQC is the proposed quantum model architecture benchmarked against classical models

# Existing Works

2. **Exploring the Potential of QEEGNet for Cross-Task and Cross-Dataset Electroencephalography Encoding with Quantum Machine Learning [6]**




| Model | Validation Accuracy | Test Accuracy |
|---|---|---|
| EEGNet | 39.86%±0.138 | 37.7%±0.126 |
| QEEGNet | **42.1%±0.142** | **38.1%±0.127** |

**Figure 3:** QEEGNet Architecture and Performance

# Existing Works

3. **Classifying Hotspots Mutations for Biosimulation with Quantum Neural Networks and Variational Quantum Eigensolver** [7]

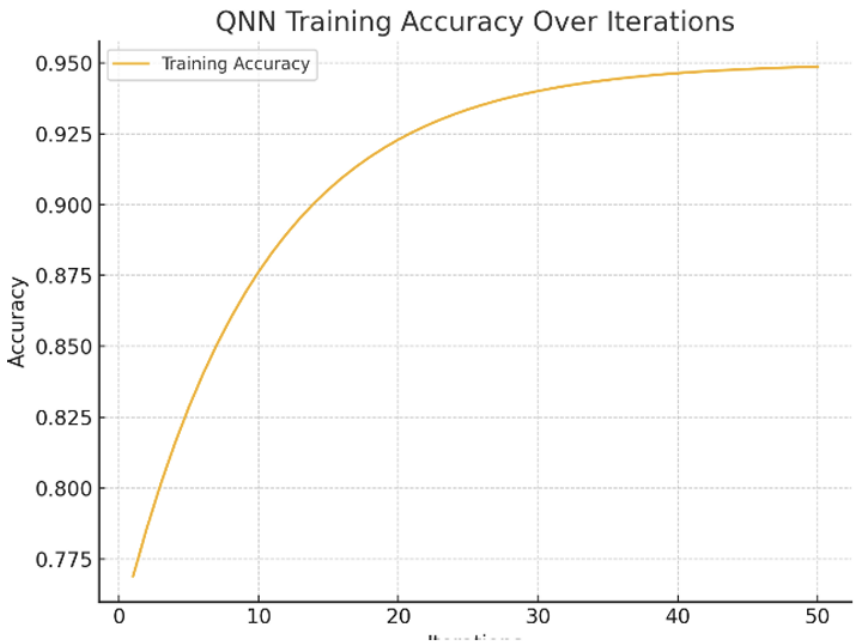



Figure 1: The accuracy of the QNN model demonstrated a steady improvement over 50 iterations, converging to a plateau near 92%. The iterative nature of training underscored the robustness of the optimization process and the model's capacity to generalize across the dataset.

# Existing Works

4. **Quantum Kernel-Based Long Short-term Memory** [8]
   - Proposed a novel LSTM architecture adding a quantum block inside RNN block reducing number of parameters

COMPARISON OF PARAMETERS FOR QK-LSTM AND LSTM NETWORKS.

| Parameter | QK-LSTM | LSTM |
|---|---|---|
| Epochs | 100 | 100 |
| Learning Rate | 0.1 | 0.1 |
| Number of Tags | 3 | 3 |
| Vocabulary Size | 5 | 5 |
| Embedding Dimension | 8 | 8 |
| Hidden Dimension | 6 | 6 |
| Number of Qubits in Quantum Kernel Circuit | 4 | – |
| **Total Trainable Parameters** | **183** | 477 |

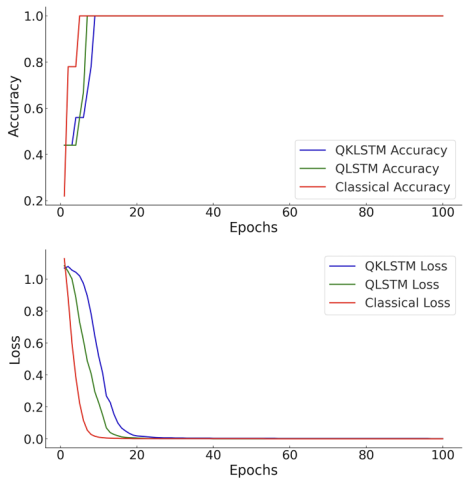



**Figure 4:** Quantum LSTM Performance Metrics

# Existing Work

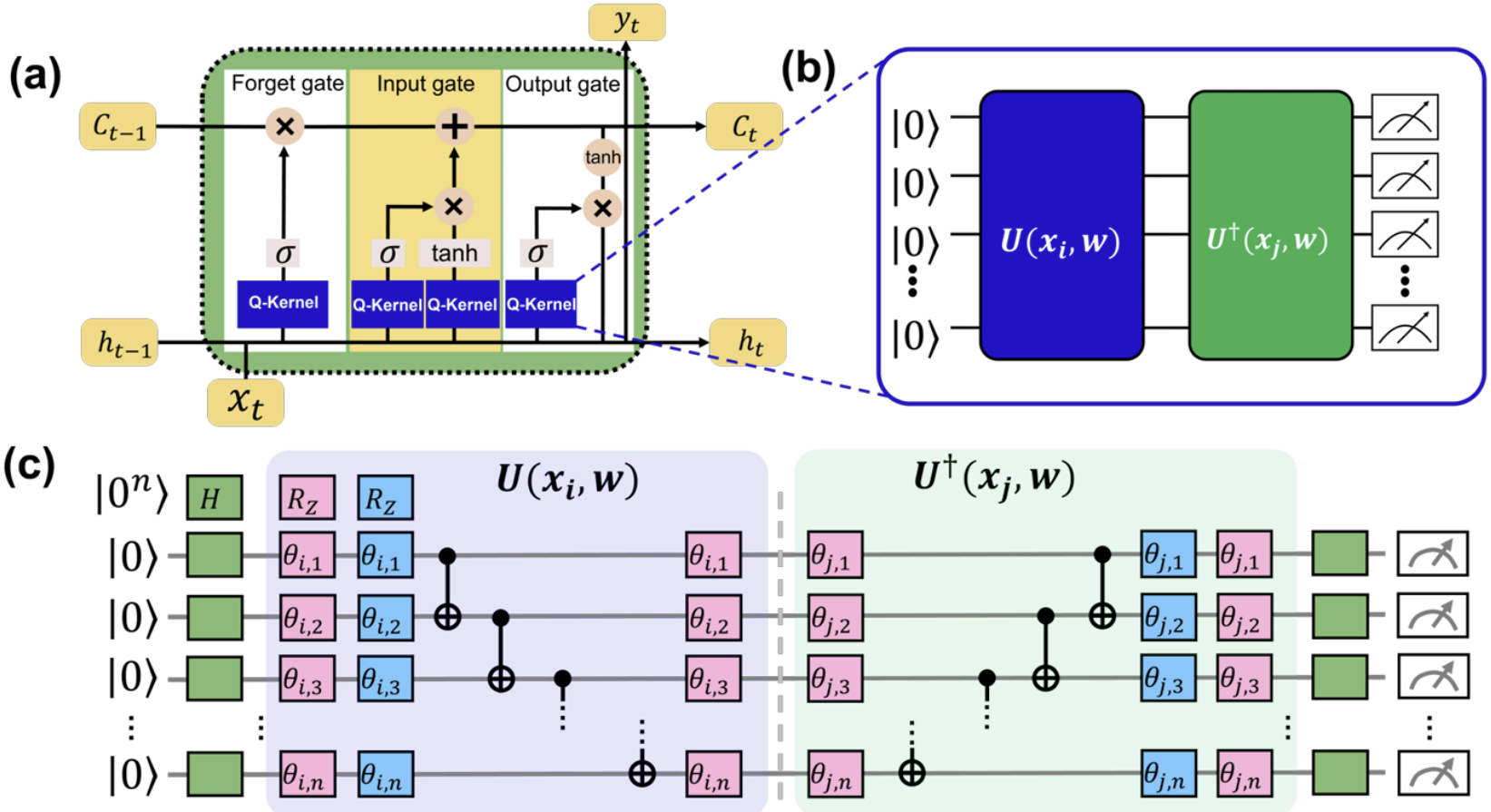



**Figure 5:** LSTM cell with quantum variational block

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# Dataset Description

# Dataset Description

| Dataset | Instances | Features | Feature Types | Task |
|---|---|---|---|---|
| **Differentiated Thyroid Cancer Recurrence** [9] | 383 | 16 | Real, Categorical, Integer | Classification |
| **Cirrhosis Patient Survival Prediction** [10] | 418 | 17 | Real, Categorical | Classification |
| **Glioma Grading Clinical and Mutation Features** [11] | 839 | 23 | Real, Categorical, Integer | Classification, Other |
| **TCGA Kidney Cancers** [12] | 1024 | 60660 | Real | Classification, Clustering |

**Table 1:** Comparison of four tabular clinical datasets for health and medicine applications where number of instances is smaller

# Dataset Description

**BEED: Bangalore EEG Epilepsy Dataset [13]**

| Instances | Features | Feature Types | Task |
|---|---|---|---|
| 8000 | 17 | Integer | Classification |

**Table 2:** Tabular clinical dataset with moderate number of instances

# Dataset Description

## CDC Diabetes Health Indicators [14]

| Instances | Features | Feature Types | Task |
|---|---|---|---|
| 253,680 | 21 | Categorical, Integer | Classification |

**Table 3:** Tabular clinical dataset with large number of instances

# Reasons for Choosing the Datasets mentioned

- **Literature gap:** No publication of QML models trained by those datasets; existing research uses classical methods (Random Forest, XGBoost, XAI)
- **Quantum encoding feasibility:**
  1. Small datasets (383 samples): Basis encoding
  2. Medium datasets (1000-5000): Angle and amplitude encoding
  3. Large datasets ($>$ 10000): Amplitude and Higher-order encoding with dimensionality reduction
- **Clinical relevance:** Well-balanced dataset suitable for demonstrating quantum advantages in healthcare

# Reasons for Choosing the Datasets mentioned

- **Ideal for benchmarking:** Tabular structure perfect for comparing quantum kernel methods and VQC against classical baselines
- **Novel contribution:** Establishes new benchmarks in quantum medical ML

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# Contribution

# Contribution

- **Quantum-Enhanced Classification:** Hybrid quantum-classical approach leveraging superposition and entanglement for high-dimensional data.
- **Improved Accuracy & Efficiency:** Expected higher classification performance with reduced computational cost compared to classical methods.
- **Robust & Generalizable:** Adaptable framework for multiple brain tumor datasets, supporting practical clinical research.
- **Advancing Quantum ML:** Demonstrates a real-world application of quantum machine learning in biomedical data analysis.

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# Project Plan

# Project Plan

- **Phase I (25%): Data Preprocessing & Encoding, Model training with low accuracy**
  - Implementation of classical normalization and quantum feature maps (e.g., Angle/Amplitude Encoding). Train one quantum, one classical model.
- **Phase II (50%): Classical Baseline Establishment**
  - Training and hyperparameter tuning of SOTA classical models (e.g., XGBoost, Random Forest).
- **Phase III (75%): Quantum Model Implementation**
  - Train QNN and Quantum Kernels on NISQ simulators.
- **Phase IV (100%): Comparative Analysis**
  - Rigorous statistical evaluation of accuracy, convergence speed, and resource overhead across all models.

# Project Plan

- **Extra : Novel quantum model architecture design**
  - *Hybrid Integration:* Embedding quantum circuits within classical deep learning pipelines.
  - *Layer Replacement:* Substituting specific classical/quantum layers/building blocks with variational quantum circuits.

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# Initial Progress

# Initial Progress

- **Literature Review:** Conducted preliminary analysis of state-of-the-art QML architectures, benchmarking methodologies, and NISQ-era hardware constraints.
- **Data Acquisition:** Collected datasets for quantum encoding.
- **Environment Configuration:** Successfully configured the quantum simulation framework using the Qiskit SDK.
- **Model Selection:** Identified candidate quantum algorithms (VQC, QSVM) and feature map strategies for comparative analysis.

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# Thank You!